CDEQ V6.0: TANXLON V-shaped Dual-Oscillator QTXS Stepping Transition
中文：琰珑 V 型双振子 QTXS 步进瞬时跃迁理论

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Abstract

All dynamic systems exhibit three fundamental constraints: (i) stable but incapable of actuation; (ii) actuated but unstable; (iii) pseudo-self-sustainment. This work establishes CDEQ V6.0 as a fixed theoretical framework and instantiates it into the TANXLON 20° V-shaped dual-oscillator system, forming a complete set of QTXS stepping transition dynamics with intrinsic convergence, controllable actuation, and mandatory safety verification.

The system strictly inherits 16 classical axioms of physics and thermodynamics, follows the topological stability mechanism of PCTT-CW (adjacent reinforce, interval restrain), is constrained by CENHE-LEX rigid law, and verified by GCVT-10 10-step safety interlock.

This paper presents the fixed, plain-text, self-consistent mathematical system for TANXLON dual oscillators. All formulas, symbols and structures are final and unchangeable. The framework achieves global stability, critical transition and reliable self-sustainment in a unified closed form.

1. Introduction

1.1 Universal Problem Statement

All dynamic systems face three core limitations:

1. Stable but non-actuated: can maintain stability but cannot actuate, accelerate or self-sustain.
2. Actuated but unstable: can actuate but tends to diverge, drift or collapse rapidly.
3. Pseudo-self-sustainment: transient stability leads to false activation and sudden failure.

The TANXLON V-shaped dual-oscillator system, as the official engineering instance of CDEQ V6.0, provides a unified framework for global dynamic stability and self-sustainment.

1.2 CDEQ Version Definition (Fixed & Unchangeable)

· CDEQ V1.0: Constructive Framework for Lyapunov Stability
· CDEQ V2.0: Constructive Lyapunov Stability & Master Equation
· CDEQ V3.0: Self-Sustained Aerothermal Propulsion
· CDEQ V3.5: Critical Transition & Self-Sustained Dynamics
· CDEQ V5.0: Global Self-Sustainment & Full Framework Closure
· CDEQ V6.0: TANXLON V-shaped Dual-Oscillator QTXS Stepping Transition

1.3 Topological Foundation: PCTT-CW

PCTT-CW (Penta-Cyclic Clockwise Topology) is the minimal and optimally stable cyclic topology.

· **Adjacent nodes:** mutual reinforce to support state evolution
· **Interval nodes:** mutual restrain to eliminate overflow and divergence

This natural equilibrium mechanism is the fixed topological foundation of TANXLON dual oscillators.

2. Bio-inspired and Engineering Prototypes

The design of TANXLON is structurally analogous and functionally isomorphic to five natural and engineering prototypes:

2.1 Pendulum Systems: Periodic Energy Storage and Release

Pendulum dynamics, structurally analogous to pirate ship and swing motion, demonstrates periodic energy storage, phase locking, and fixed-point release — the principle origin of resonance dynamics in this work.

2.2 Athens "FREAK" 20° Dual Oscillator

The 20° V-shape is mechanically isomorphic to precision dual-oscillator design, exemplified by the Athens "FREAK" watch system. This structure provides optimal decoupling and synchronization — the topological origin of the V-configuration.

The V-shaped configuration is a proven stable topology in aeronautical engineering. It provides inherent structural symmetry, dynamic balance, and mechanical efficiency — advantages universally recognized in flying wing designs and fully aligned with the requirements of the TANXLON system. Unlike conventional atmospheric vehicles designed for sustained cruising, this system is intended for discrete stepping, rapid transition, and impulsive transient propulsion in space-like environments.

2.3 Mantis Shrimp 81% Energy Conversion Efficiency

Elastic-to-kinetic conversion efficiency $\eta = 0.81$ (81%) is experimentally validated in mantis shrimp biomechanics. This value represents the fraction of stored elastic energy converted to kinetic energy in a single strike, with the remainder dissipated as heat, sound, and structural vibration. It is fixed as the high-efficiency transformation benchmark for TANXLON, emphasizing one-way conversion, not cyclical recovery. Once released, kinetic energy is not recaptured; the system resets to reload.

2.4 Qian Xuesen “Skipping Stone” Trajectory

Intermittent stepping motion is kinematically analogous to Qian Xuesen’s “skipping stone” trajectory. As a general principle, intermittent motion can be more stable and efficient than continuous actuation — the conceptual origin of QTXS stepping transition.

2.5 Space Kangaroo (Dual-leg Impulse Configuration)

Dual-leg energy storage, synchronous actuation, and balanced posture are functionally isomorphic to kangaroo

locomotion. This serves as the morphological analogy for TANXLON's balanced actuation structure.

These five principles converge into one fixed system:
TANXLON V-shaped dual-oscillator QTXS stepping transition.

3. Key Physical Anchors

· **Base period:** 167 ms (empirically selected fundamental frequency for resonance optimization)
· **Energy conversion efficiency:** 81% (experimentally validated, fixed value)
· **V-angle:** 20° (mechanically analogous to optimal decoupling angle in precision systems)
· **Verification window:** GCVT-10 (fixed 10-step safety interlock)

4. Axiom System

4.1 Thirteen Fundamental Axioms

1. Energy Conservation (First Law of Thermodynamics)
2. Entropy Increase Principle (Second Law of Thermodynamics)
3. Third Law of Thermodynamics
4. Zeroth Law of Thermodynamics
5. Newton's First Law (Inertia)
6. Newton's Second Law (Force–Motion Relation)
7. Newton's Third Law (Action–Reaction)
8. Conservation of Momentum
9. Conservation of Mass
10. Ideal Gas Law (PV = nRT)
11. Lyapunov Stability Principles
12. Feedback Control Theory Principles
13. Mathematical Convergence and Limit Axioms

4.2 Three Boundary Axioms (Fixed Rules)

1. **Initial external energy input is required (no perpetual motion).**
(Metaphor: Mother gives 10 yuan; without it, Xiao Ming starves.)
2. **Self-sustained operation maintains momentum through inertial glide and gravitational assist, not through energy recovery or cyclic regeneration.**
The system does not recapture energy after release; it relies on vacuum inertia and natural gravitational fields for prolonged flight, with all energy conversions being strictly one-way.
3. **Mode transition requires GCVT verification for 10 consecutive steps.**
(The swing must complete 10 stable oscillations before release.)

The TANXLON system strictly and fully obeys all 16 axioms without exception.

5. Core Constraint & Verification Units

5.1 CENHE-LEX: Rigid Constraint Unit (Fixed)

CENHE-LEX is the highest-priority stability enforcer:

1. Stabilize the central state to eliminate drift
2. Maintain structural conservation
3. Restrain overload and enforce state return to safe bounds
4. Forbid unstable or non-compliant operation

5.2 GCVT-10: Global Convergence Verification (Fixed)

GCVT-10 is the mandatory 10-step consecutive safety interlock.
Convergence conditions (fixed, must hold ∀10):

1. Central stability: $|E_{11} - 5| < \varepsilon$
2. Structural balance: $|\Sigma - 15| < \varepsilon$
3. Cost threshold: $\Phi < \Gamma$

GCVT-10 eliminates pseudo-stability and false activation by design.

6. CDEQ V6.0 Core Mathematical System

6.1 V2.0 Basic Brake Formula (Fixed)

$$X' = \text{clip}[\ \varphi X_{-1} - \varphi \bar{X}_{-2} + \omega\xi\ ]$$

6.2 V6.0 Ultimate Harpoon Equation (Fixed, Unchangeable, Core Result)

$$X' = \text{clip}[\ \varphi X_{-1} - \varphi \bar{X}_{-2} + \omega\xi\ ] + (X \geq X_th \wedge P_self \geq P_ext)\cdot\nabla C$$

Unified global constraint (Fixed):
$\Phi < \Gamma\ \&\ |E_{11}-5| < \varepsilon\ \&\ |\Sigma-15| < \varepsilon\ \forall 10$

6.3 Core Modules (Fixed Definition)

1. clip[·] = V2.0 Brake Module (continuously active)
2. $(X \geq X_th \wedge P_self \geq P_ext)$ = V3.5 Throttle Switch
3. ∇C = V3.5 Throttle Drive (engineering assumption, see Section 9)
4. ∀10 constraint = GCVT-10 Mandatory Verification

7. TANXLON 20° V-shaped Dual Oscillator Equations

7.1 Left Oscillator

$$\begin{aligned} X'_L = \text{clip}[\ & \varphi X_L(-1) - \varphi \bar{X}_L(-2) + \omega W_L \xi_L\ ] \\ & + C_V[0,0]\cdot P_self_L + C_V[0,1]\cdot P_self_R \\ & + I(\ |\varphi_L - \varphi_R| < \Delta\varphi_thresh \wedge \langle X\rangle_avg \geq X_th\ )\ \cdot\ \nabla C \end{aligned}$$

Note: ∇C is an engineering assumption (see Section 9).

7.2 Right Oscillator

$$
\begin{aligned}
X'_R = \mathrm{clip}[\ & \varphi X_R(-1) - \varphi X_R(-2) + \omega W_R \xi_R\ ] \\
& + C_V[1,0] \cdot P_{self_L} + C_V[1,1] \cdot P_{self_R} \\
& + I(\ |\varphi_L - \varphi_R| < \Delta\varphi_{thresh} \wedge \langle X \rangle_{avg} \geq X_{th}\ ) \cdot \nabla C
\end{aligned}
$$

Note: ∇C is an engineering assumption (see Section 9).

7.3 20° V-coupling Matrix (Fixed)

$$
\begin{aligned}
C_V = [& \\
& [\cos 20°, \sin 20°], \\
& [\sin 20°, \cos 20°] \\
] & \\
\approx [& \\
& [0.9397, 0.3420], \\
& [0.3420, 0.9397] \\
] &
\end{aligned}
$$

7.4 Average State (Fixed)

$$\langle X \rangle_{avg} = (X_L + X_R) / 2$$

7.5 GCVT-10 for Dual Oscillators (Fixed)

$$|\varphi_L - \varphi_R| < \Delta\varphi_{thresh}\ \forall 10$$
$$\langle X \rangle_{avg} \geq X_{th}\ \forall 10$$

8. Segmented State Machine

- **$X' = 0$,** Condition: $X < X_{th} \wedge \neg GCVT$
  [Unqualified · Inhibited State]
- **$X' = \mathrm{clip}[...] + \nabla C_{sub}$,** Condition: $X \geq X_{th} \wedge GCVT \wedge P_{self} < P_{ext}$
  [Brake + Subcritical Actuation Mode]
- **$X' = \mathrm{clip}[...] + \nabla C_{crit}$,** Condition: $X \geq X_{th} \wedge GCVT \wedge P_{self} \geq P_{ext} \wedge \neg SS$
  [Brake + Critical Actuation Mode]
- **$X' = \mathrm{clip}[...] + \nabla C_{sust}$,** Condition: $X \geq X_{th} \wedge GCVT \wedge P_{self} \geq P_{ext} \wedge SS$
  [Brake + Fully Self-Sustained Actuation Mode]

Unified constraint: GCVT-10 $\forall 10$ (Fixed)

9. Engineering Implementation & Disclaimer

DISCLAIMER: All formulas, structures and implementations are for theoretical demonstration and proof-of-concept only. They do NOT guarantee engineering accuracy, reliability, certification or real-world performance. ∇C is an engineering assumption, not an experimentally validated physical fact.

10. Paradigm & Academic Record

CDEQ V6.0 and TANXLON represent a fixed, closed paradigm for dynamic stability, critical transition and global self-sustainment.

All intellectual content is asserted by Jiang Wenjia.
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11. Conclusion

CDEQ V6.0 TANXLON V-shaped dual-oscillator QTXS stepping transition is the fixed official engineering instance of the complete CDEQ paradigm. It integrates: biomechanical analogies, precision mechanical isomorphism, resonance principles, kinematic analogy, dual-leg functional analogy, PCTT-CW topology, CENHE-LEX constraints, V2.0 brake, V3.5 throttle, and GCVT-10 safety interlock.

The system is stable, controllable, self-sustained (via inertia and gravity, not energy recovery), fully compliant with classical physics, and formally finalized.

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